

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Use the atomic masses given in each question. A periodic table and the answer key are at the back.

Parts 1-2 · Ionic bonds and ions

Q1 IC1-LO1

An ionic bond is best described as:

- A. a shared pair of electrons
- B. the attraction between oppositely charged ions
- C. the attraction between two neutral atoms
- D. the repulsion between two cations

Q2 IC1-LO1

A magnesium atom (Group 2A) reaches an octet by losing two electrons.

True / False

Q3 IC1-LO2

Write the ion formed by sulfur (Group 6A), including its charge. Type it on one line, e.g. Cl⁻ for Cl⁻.

Q4 IC1-LO2

Which ion is **isoelectronic** with argon?

- A. Na^+
- B. K^+
- C. Mg^{2+}
- D. F^-

Q5 IC1-LO2

Predict the usual charge of each element's ion.

Options: $1+$ · $2+$ · $3+$ · $1-$ · $2-$ · $3-$

Al (3A)

Ba (2A)

N (5A)

I (7A)

Part 3 · Formulas of ionic compounds

Q6 IC1-LO3

Write the formula of the compound of Na^+ and O^{2-} .

Q7 IC1-LO3

Write the formula of the compound of Ca^{2+} and NO_3^- .

Q8 IC1-LO3

Write the formula of the compound of Al^{3+} and SO_4^{2-} .

Q9 IC1-LO3

Which compound is ionic?

- A. PCl_3
- B. CO_2
- C. NH_4Cl
- D. SO_2

Part 4 • Naming

Q10 IC1-LO4

Give the Stock name of CuCl_2 .

Q11 IC1-LO4

What is the formula of ferric oxide?

- A. FeO
- B. Fe_2O_3
- C. Fe_3O_2
- D. FeO_3

Q12 IC1-LO4

Match each polyatomic ion to its name.

Options: nitrate · nitrite · sulfate · sulfite · bicarbonate (hydrogen carbonate) · hydroxide

NO_2^-

SO_3^{2-}

HCO_3^-

OH^-

Q13 IC1-LO4

Stannous fluoride is added to some toothpastes. Its formula is:

- A. SnF
- B. SnF_2
- C. SnF_4
- D. Sn_2F

Part 5 • Formula mass

Q14 IC1-LO5

Potassium chloride is used as a salt substitute. Calculate its formula mass ($\text{K} = 39.10$, $\text{Cl} = 35.45$).

Q15 IC1-LO5

Calculate the formula mass of gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ($\text{Ca} = 40.08$, $\text{S} = 32.07$, $\text{O} = 16.00$, $\text{H} = 1.01$).

Part 6 • Properties of ionic compounds

Q16 IC1-LO6

Which of these conducts electricity?

- A. solid NaCl
- B. distilled water
- C. NaCl dissolved in water
- D. dry sugar crystals

Q17 IC1-LO6

Ionic compounds are brittle because shifting a layer of ions brings like charges together, and they repel.

True / False

Q18 IC1-LO6

Write the dissociation equation for potassium sulfate, K_2SO_4 , dissolving in water. Include state symbols.

Pharmacy scenario · Iron tablets

Scenario

A patient with iron-deficiency anaemia is prescribed **ferrous sulfate** tablets. Use your knowledge of ions to answer the next four questions (Fe = 55.84, S = 32.07, O = 16.00).

Q19 IC1-LO4

Write the formula of ferrous sulfate.

Q20 IC1-LO4

Give the Stock name of ferrous sulfate.

Q21 IC1-LO5

Calculate the formula mass of ferrous sulfate.

Q22 **IC1-LO6**

The tablet dissolves in the stomach. Write the dissociation equation, and state which ion the body absorbs to make haemoglobin.

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 25 minutes. Questions follow the six parts of the lecture and end with a pharmacy scenario.

Answer key

Parts 1-2 · Ionic bonds and ions

Q1 IC1-LO1

An ionic bond is best described as:

- A. a shared pair of electrons
- B. the attraction between oppositely charged ions**
- C. the attraction between two neutral atoms
- D. the repulsion between two cations

Electrons are transferred, and the resulting cations and anions attract each other.

Q2 IC1-LO1

A magnesium atom (Group 2A) reaches an octet by losing two electrons.

True / False → **True**



Q3 IC1-LO2

Write the ion formed by sulfur (Group 6A), including its charge. Type it on one line, e.g. Cl⁻ for Cl⁻.

S²⁻

Sulfur gains two electrons. Write the number before the sign: S²⁻.

Q4 IC1-LO2

Which ion is **isoelectronic** with argon?

A. Na^+

B. K^+

C. Mg^{2+}

D. F^-

K^+ has 18 electrons, like Ar. Na^+ , Mg^{2+} and F^- have 10 (like Ne).

Q5 IC1-LO2

Predict the usual charge of each element's ion.

Al (3A) $\rightarrow$ 3+; Ba (2A) $\rightarrow$ 2+; N (5A) $\rightarrow$ 3-; I (7A) $\rightarrow$ 1-

Charge follows the group: 1A 1+, 2A 2+, 3A 3+, 5A 3-, 6A 2-, 7A 1-.

Part 3 • Formulas of ionic compounds

Q6 IC1-LO3

Write the formula of the compound of Na^+ and O^{2-} .

Na_2O

Two Na^+ (2+) balance one O^{2-} (2-).

Q7 IC1-LO3

Write the formula of the compound of Ca^{2+} and NO_3^- .

$\text{Ca}(\text{NO}_3)_2$

Two nitrate ions are needed; a polyatomic ion goes in brackets.

Q8 IC1-LO3

Write the formula of the compound of Al^{3+} and SO_4^{2-} .

$\text{Al}_2(\text{SO}_4)_3$

LCM of 3 and 2 is 6: two Al^{3+} (6+) and three SO_4^{2-} (6-).

Q9 IC1-LO3

Which compound is ionic?

- A. PCl_3
- B. CO_2
- C. NH_4Cl**
- D. SO_2

NH_4^+ is the ammonium ion, so NH_4Cl is ionic. The others contain only nonmetals and no polyatomic ion.

Part 4 • Naming

Q10 IC1-LO4

Give the Stock name of CuCl_2 .

copper(II) chloride (cupric chloride)

Two Cl^- means copper is 2+.

Q11 IC1-LO4

What is the formula of ferric oxide?

- A. FeO
- B. Fe_2O_3**
- C. Fe_3O_2
- D. FeO_3

Ferric = Fe^{3+} (the higher charge). Two Fe^{3+} balance three O^{2-} .

Q12 IC1-LO4

Match each polyatomic ion to its name.

$\text{NO}_2^- \rightarrow$ nitrite; $\text{SO}_3^{2-} \rightarrow$ sulfite; $\text{HCO}_3^- \rightarrow$ bicarbonate (hydrogen carbonate); $\text{OH}^- \rightarrow$ hydroxide

The -ate ion has one more oxygen than the -ite ion.

Q13 IC1-LO4

Stannous fluoride is added to some toothpastes. Its formula is:

- A. SnF
- B. SnF₂**
- C. SnF₄
- D. Sn₂F

Stannous = Sn²⁺ (the lower tin charge), so two F⁻ are needed.

Part 5 • Formula mass

Q14 IC1-LO5

Potassium chloride is used as a salt substitute. Calculate its formula mass (K = 39.10, Cl = 35.45).

74.55 amu

$39.10 + 35.45 = 74.55$ amu.

Q15 IC1-LO5

Calculate the formula mass of gypsum, CaSO₄·2H₂O (Ca = 40.08, S = 32.07, O = 16.00, H = 1.01).

172.19 amu

$\text{CaSO}_4 = 40.08 + 32.07 + 64.00 = 136.15$; $2\text{H}_2\text{O} = 2 \times 18.02 = 36.04$; total 172.19 amu.

Part 6 • Properties of ionic compounds

Q16 IC1-LO6

Which of these conducts electricity?

- A. solid NaCl
- B. distilled water
- C. NaCl dissolved in water**
- D. dry sugar crystals

In solution the ions are free to move and carry the current.

Q17 IC1-LO6

Ionic compounds are brittle because shifting a layer of ions brings like charges together, and they repel.

True / False → **True**

That repulsion makes the crystal shatter along a plane.

Q18 IC1-LO6

Write the dissociation equation for potassium sulfate, K_2SO_4 , dissolving in water. Include state symbols.

$K_2SO_4(s) \rightarrow 2K^+(aq) + SO_4^{2-}(aq)$. The sulfate ion stays together as one polyatomic ion, and the equation must show two K^+ ions.

Pharmacy scenario · Iron tablets

Q19 IC1-LO4

Write the formula of ferrous sulfate.

$FeSO_4$

Ferrous = Fe^{2+} , which balances one SO_4^{2-} .

Q20 IC1-LO4

Give the Stock name of ferrous sulfate.

iron(II) sulfate

The Roman numeral shows the 2+ charge on iron.

Q21 IC1-LO5

Calculate the formula mass of ferrous sulfate.

151.91 amu

$55.84 + 32.07 + 4 \times 16.00 = 151.91$ amu.

Q22 IC1-LO6

The tablet dissolves in the stomach. Write the dissociation equation, and state which ion the body absorbs to make haemoglobin.

$\text{FeSO}_4(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$. The iron is absorbed as the iron(II) ion, Fe^{2+} . Dissolving and dissociating into ions is why many medicines are given as ionic salts.

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